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A NUMERICAL APPROACH TO THE COMPLETE STRESS-STRAIN CURVE OF CONCRETE

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ABSTRACT

This paper presents the experimental justification of two previously published formulas, Eqs. 2) and 6), for the estimation of the complete stress-strain diagram of concrete. Eq. 2) combined with Eq. 3) differs from the other formulas offered in the literature for similar purpose in that provides more relative curvature in the diagram for concretes of lower strengths. Also, with Eq. 6), it can take the fact into consideration that the value of ε_o increases with increasing concrete strength. The result of these refinements is that the stress-strain diagrams calculated by these formulas fit better the experimentally obtained diagrams and within wider limits than the similar formulas available in the literature. (Figs. 5a through 5d, 8a through 8d, and 9.)

Diese Arbeit veranschaulicht die experimentelle Rechtfertigung zweier früher veröffentlichter Formeln, Gleichungen 2) und 6) zur Bestimmung des vollständigen Spannungs-Dehnungs-Diagramms des Betons. Die mit Gleichung 2 verbundene Gleichung 3 weicht insofern von den anderen, bisher in der Literatur bekannten Formeln der gelichen Richtung ab, als sie mehr relative Krümmungen in dem Diagramm für Beton niederer Festigkeit aufzeit. Auch kann man mit Gleichung 6 annehmen, daß der Wert ε_o mit zunehmender Beton festigkeit zunimmt. Das Resultat dieser Verbesserungen zeigt, daß die mit diesen Formeln berechneten Spannungs-Dehnungs-Diagramme sich besser und im weiteren Rahmen den experimentell erhaltenen angleichen, als ähnliche Formeln, die man der Literatur entnehmen kann.

Introduction

This paper is the continuation of the previous work of the writer on the stress-strain diagram. (1)-(3) The purpose of this study is (a) to present formulas for the estimation of the complete stress-strain diagram of normal-weight concrete, made with a given aggregate and tested with a given procedure, under short-term loading either solely from the f_o compressive strength, or from the combination of the compressive strength and the measured ε_o unit strain in concrete at the f_o ultimate stress (Fig. 1); and (b) to show the applicability of these formulas by demonstrating their goodness of fit to pertinent experimental results.

The discussion of the stress-strain curve of concrete is timely from a theoretical point of view because deformations may provide indirect information concerning the internal structure as well as the failure mechanism of concrete. From a practical standpoint, the ultimate-strength design of reinforced concrete elements brought the stress-strain relationship into focus. Also, a knowledge of the deformability of concrete is necessary to compute deflections of structures, to compute stresses from observed strains, to design sections of highway slabs, to compute loss prestress in prestressed elements, etc. (4) In the study of models of concrete structures, it is also necessary to know the stress-strain characteristics of the model material so that dimensional similarity may be obtained.

Stress-Strain Diagram for Compression

The diagram is influenced considerably by the testing conditions (type of the testing machine, rate and duration of loading, size and shape of the specimen, size and location of the strain gages, number of load repetitions, etc.) as well as by the age and composition of concrete, especially by the type and quantity of aggregate and by the porosity, in a similar but not identical way as the concrete strength is influenced by most of these factors. (2)

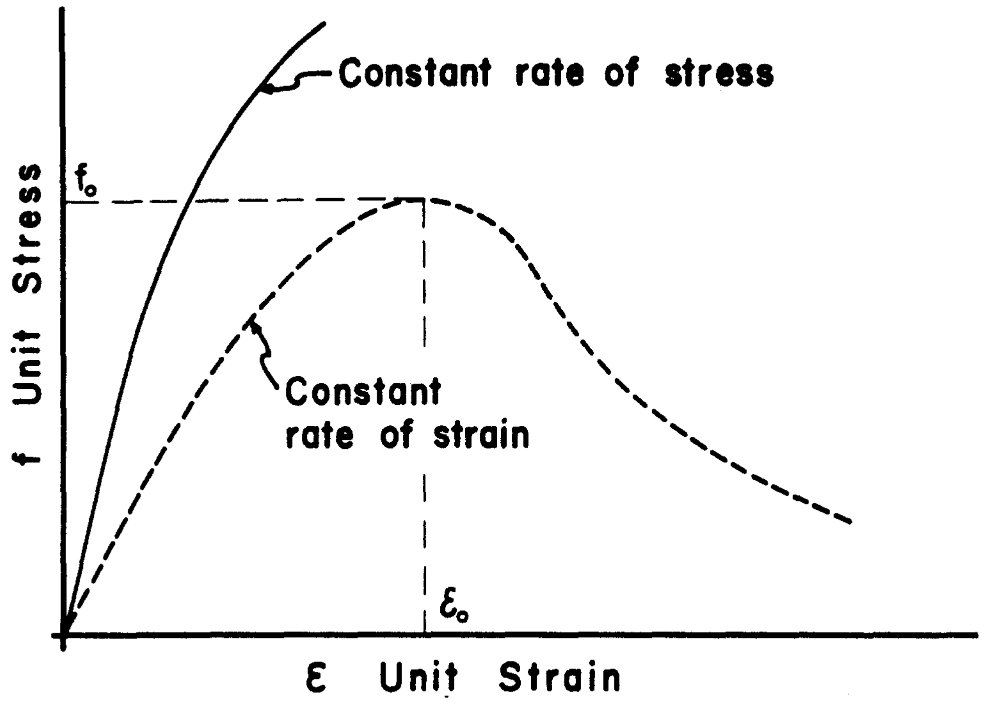


Fig. 1. Two typical stress-strain curves for concrete under uniaxial load. The top curve is characteristic of a loading process where the rate of stress increase is kept constant during the testing. The bottom curve is obtained by keeping the rate of strain increase constant.

Another interesting fact is that the stress-strain diagrams for stones and hardened cement pastes under uni-axial loading are practically straight lines almost up to the ultimate stress. Yet the same diagram is curved for mortars consisting essentially of the same two components, and even more curved for concretes (Figs. 2 and 3), as has been pointed out by Gilkey and Murphy (5). This paradox can be explained in qualitative terms by the internal cracking (7) and creep of the hardened paste in concrete under load that are produced by the stress, especially by the stress concentrations, resulting from the embedded aggregate particles (8). However, the exact nature of this problem is so complex that only empirical formulas are available in the literature for numerical approximation of the stress-strain diagram. These formulas have been discussed elsewhere. (1)

In most of these formulas the E/E_0 ratio is a fixed number regardless of the concrete strength. This restricts the limits of validity of these

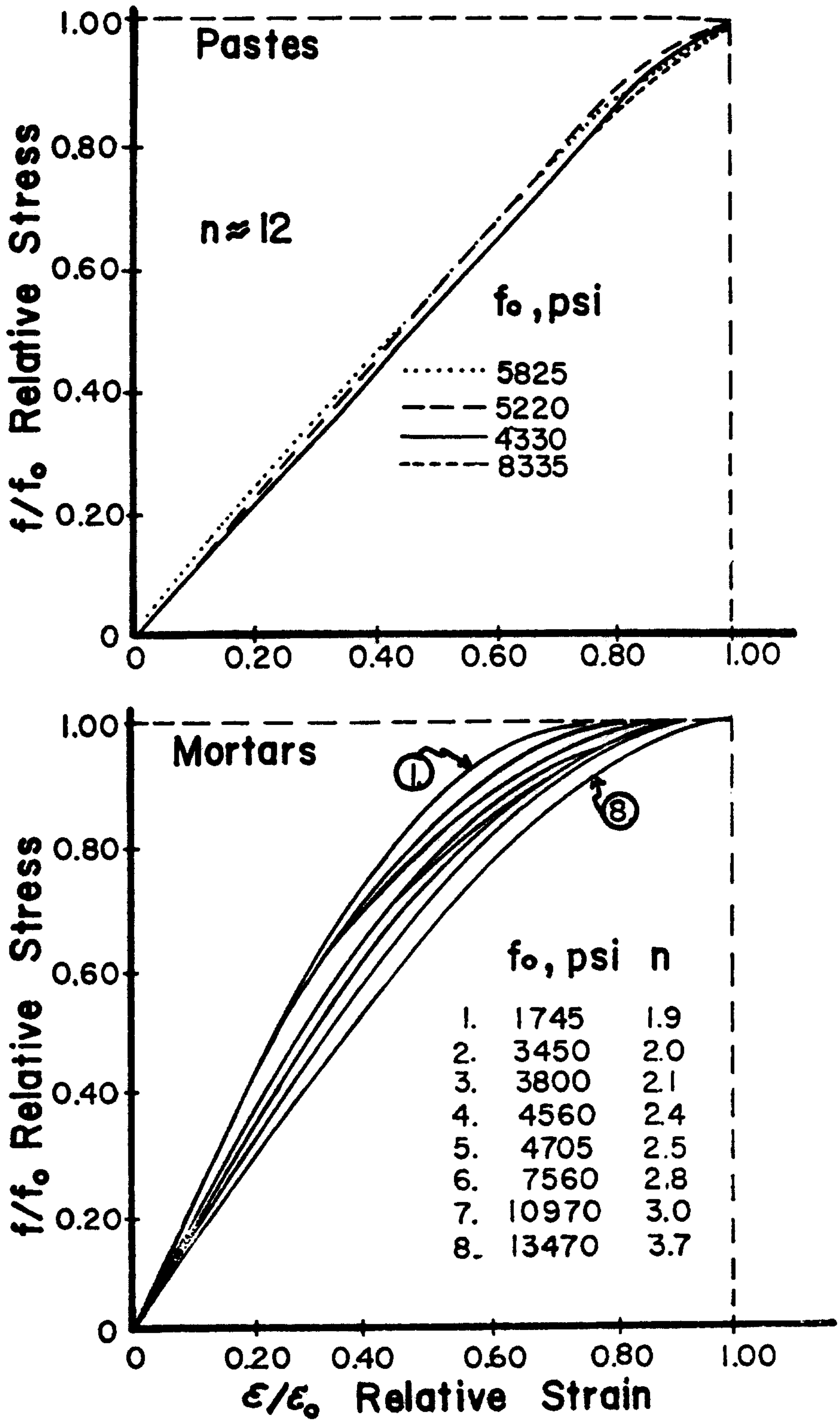


Fig. 2. Relative stress-strain diagrams for hardened pastes and mortars of various strengths along with the best fit values of n. The curves were published by Gilkey and Murphy. (5)

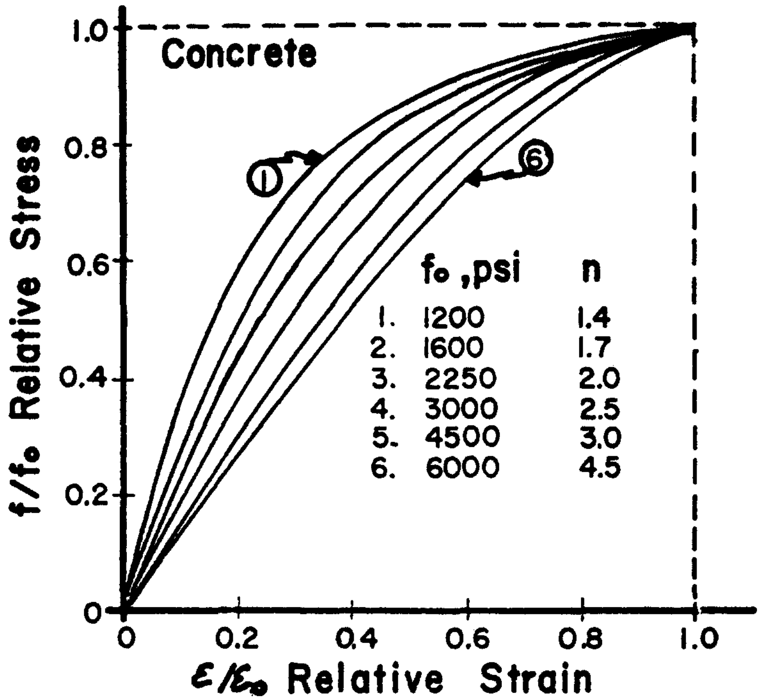


Fig. 3. Relative stress-strain diagrams for concretes of various compressive strengths along with the best fit values of n. The curves were published by Rüsch. (6)

formulas because experimental data show that the E/E_0 ratio varies from near 4 for normal concretes of 1,000 psi to about 1.3 for concretes of 10,000 psi. Consequently, when such a formula fits a concrete of medium strength, it will over estimate the stress for a given strain in the ascending branch of the stress-strain diagram for the high-strength concretes and under estimate it for low-strength concretes. (9) Formulas with variable E/E_0 ratios are more flexible because they can take the composition of the concrete into consideration directly or indirectly, such as through the concrete strength. Such a formula is the following (3):

$$f = E\varepsilon \frac{n - 1}{n - 1 + (\varepsilon/\varepsilon_0)^n} \qquad 1)$$

or, considering that at $\varepsilon = \varepsilon_0$, $E = (f_0/\varepsilon_0)n/(n - 1)$,

$$f = f_0 \frac{\varepsilon}{\varepsilon_0} \frac{n}{n - 1 + (\varepsilon/\varepsilon_0)^n} \qquad 2)$$

where the n power can be expressed as an approximate function of the compressive strength of normal-weight concrete as follows:

$$n_{concrete} = 0.4\text{x}10^{-3}\ f_o + 1.0 \qquad 3)$$

Similar formulas for cement mortars and pastes are:

$$n_{mortar} = 0.15\text{x}10^{-3}\ f_o + 1.5 \qquad 4)$$

and

$$n_{paste} = 12. \qquad 5)$$

The long fractions in Eqs. 1) and 2) represent the deviation from the linear elasticity. This might be utilized in the future for the analysis of crack propagation in concrete. Note also that these equations, combined with Eq. 3), differ from the other formulas offered in the literature for similar purpose in that they provide more relative curvature in the diagram for concretes of lower strengths.

Figure 4 illustrates Eq. 2) in relative terms for normal-weight concretes and pastes. It is impossible to produce a single formula similar to Eq. 3) for lightweight concretes in general because the various types of such concretes have greatly differing deformabilities. For instance, gas concretes show nearly linear elasticity while concretes made with expanded blast-furnace slag as aggregate have highly curved stress-strain diagrams. (10) It should also be pointed out that the formulas above are valid only for standard concrete specimens with a height-width ratio not less than two, and when the uniaxial compressive load is a short-term load which is applied at a rate that produces constant rate of strain in the specimen. (Fig. 1)

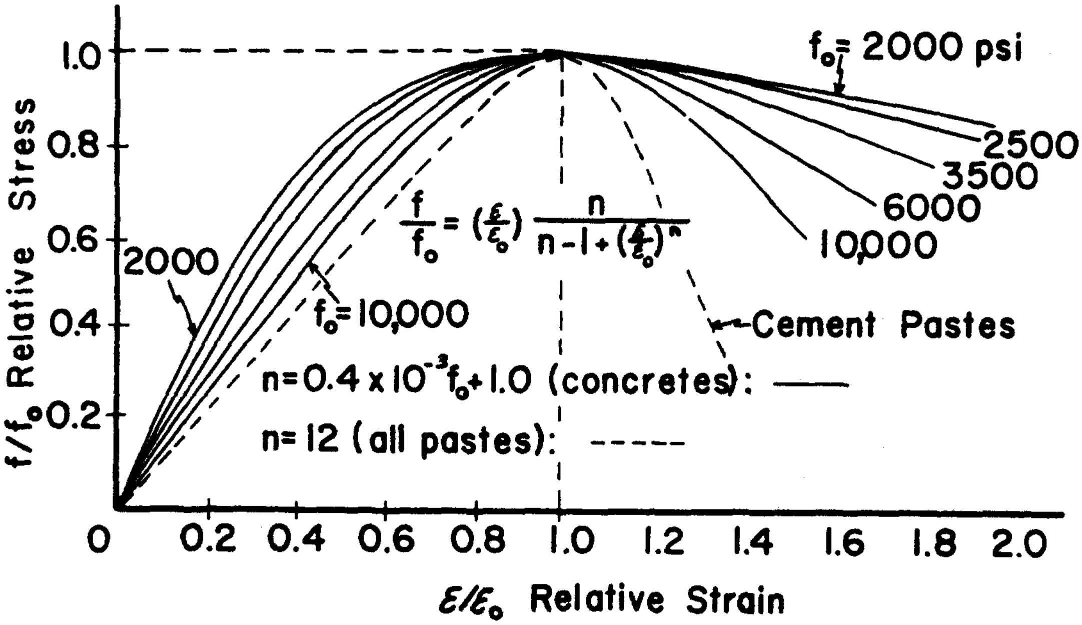
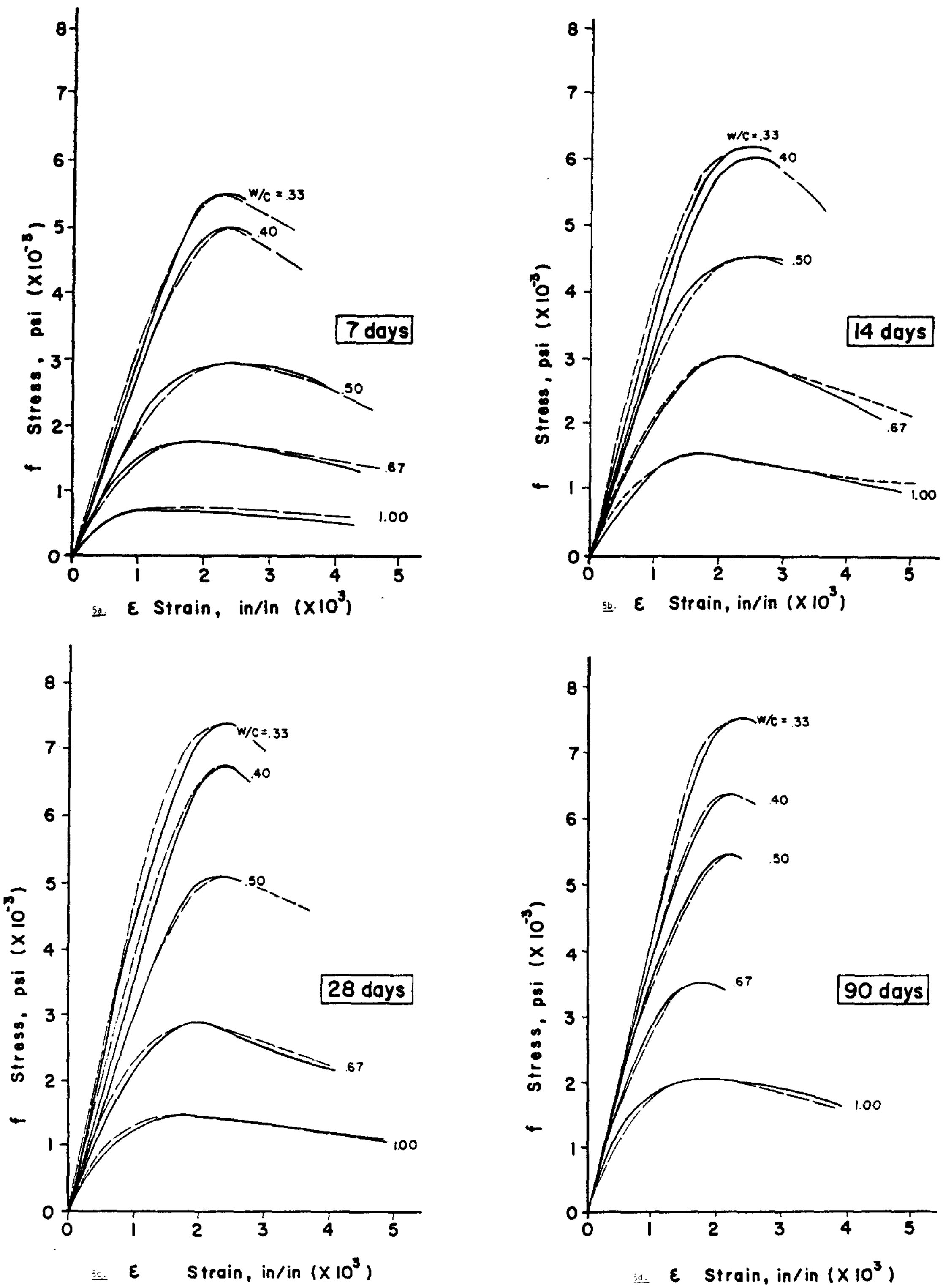


Fig. 4. Calculated relative stress-strain diagrams for normal-weight concretes of various compressive strengths and for cement pastes.

Comparison to Experimental Data

It has been shown in qualitative terms (3) that Eqs. 2) through 5) are more suitable for the description of the stress-strain diagram of a concrete than the formulas with fixed E/E_o ratio. However, it has not been examined yet that to what extent experimental data support these formulas. This examination is presented below for two different situations: (a) when both the f_o and the ε_o values are available; and (b) the more practical case, when only the value of f_o is available. Incidentally, Eq. 1) is not recommended for practical purposes, although it is an interesting formula because, for a given n value, it expresses the stress values purely in terms of deformations.

In the first situation, that is, when both f_o and ε_o are measured for a concrete, mortar, or paste, the corresponding value of n can be calculated from one of Eqs. 3) through 5), and used with Eq. 2). Since Eqs. 3) through 5) were obtained by fitting the curves of Figure 4 to those in Figures 2 and 3, the stress-strain diagrams calculated in this way obviously fit the pertinent experimental data by Gilkey (5), and by Rüsch (6). Other



Figs. 5a through 5d. Comparison of experimental stress-strain diagrams of various concretes (continuous lines) to those plotted from Eq. 2) with Eq. 3) (dash lines). The experimental curves were taken from Reference (11).

Legend

The complete curve was obtained experimentally.

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$$f = f_o \frac{\varepsilon}{\varepsilon_o} \frac{n}{n-1+(\varepsilon/\varepsilon_o)^n}$$

$$n = 0.4 \times 10^{-3} f_o + 1$$

f_o and ε_o were obtained experimentally.

—— o ——

$$f = f_o \frac{\varepsilon}{\varepsilon_o} \frac{n}{n-1+(\varepsilon/\varepsilon_o)^n}$$

$$n = 0.4 \times 10^{-3} f_o + 1$$

$$\varepsilon_o = 2.7 \times 10^{-4} \sqrt[4]{f_o}$$

f_o was obtained experimentally.

curves calculated with Eqs. 2) and 3) are compared in Figures 5a through 5d with experimental curves published by Hognestad et. al. (11) The comparison is done with 20 pairs of diagrams representing five water-cement ratios and four ages. It can be seen that despite the wide ranges in water-cement ratio (from 0.33 to 1.0 by weight) and age at testing (from 7 to 90 days), the calculated diagrams fit the experimental curves quite well.

In the second situation, when only f_o is given, the value of ε_o can be estimated from one of the available formulas in the literature, and then this value used with Eq. 2) again, as mentioned above. A convenient form of such a formula was presented earlier (3) from certain considerations, but without experimental justification, as follows:

$$\varepsilon_o = k \times 10^{-4} \sqrt[4]{f_o} \qquad 6)$$

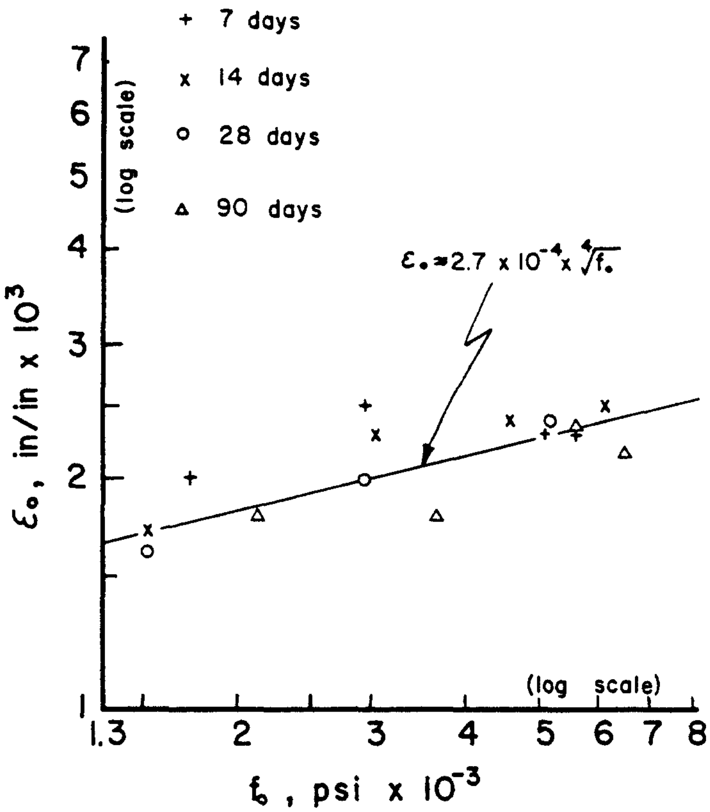


Fig. 6. Example for the relationship between ε_o and f_o for normal-weight concretes of various ages. The experimental data were taken from a paper by Hognestad et al. (11).

where k is a function of the type of mineral aggregate used and the applied test method. As can be seen from Figure 6, Eq. 6) is supported quite well by the test results on normal-weight concretes that were published by Hognestad et al. (11), (although the fit could be improved by including the age as an extra variable). Figure 7 shows that test results published by Watanabe (12) again support Eq. 6). The difference between the deformabilities of lightweight and normal-weight concretes is also illustrated in this figure.

Returning to the original question, Figures 8a through 8d demonstrate the goodness of fit of the curves calculated from the pertinent combination of Eqs. 2), 3) and 6) (the latter with k = 2.7 $in^{\frac{1}{2}}/lb^{\frac{1}{4}}$) to the same experimental curves that were discussed in connection with Figures 5a through 5d. Figure 9 shows another comparison, with k = 2.2 $in^{\frac{1}{2}}/lb^{\frac{1}{4}}$, to a set of

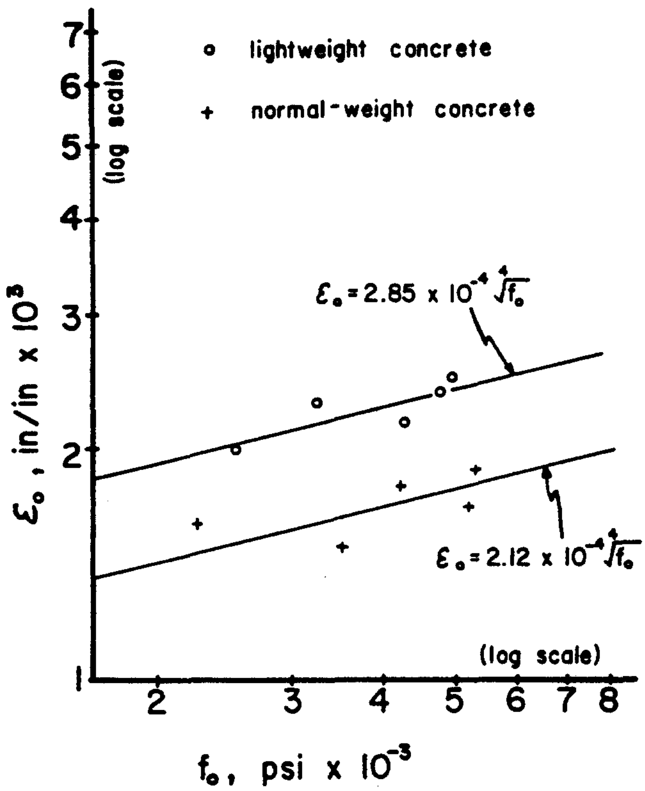
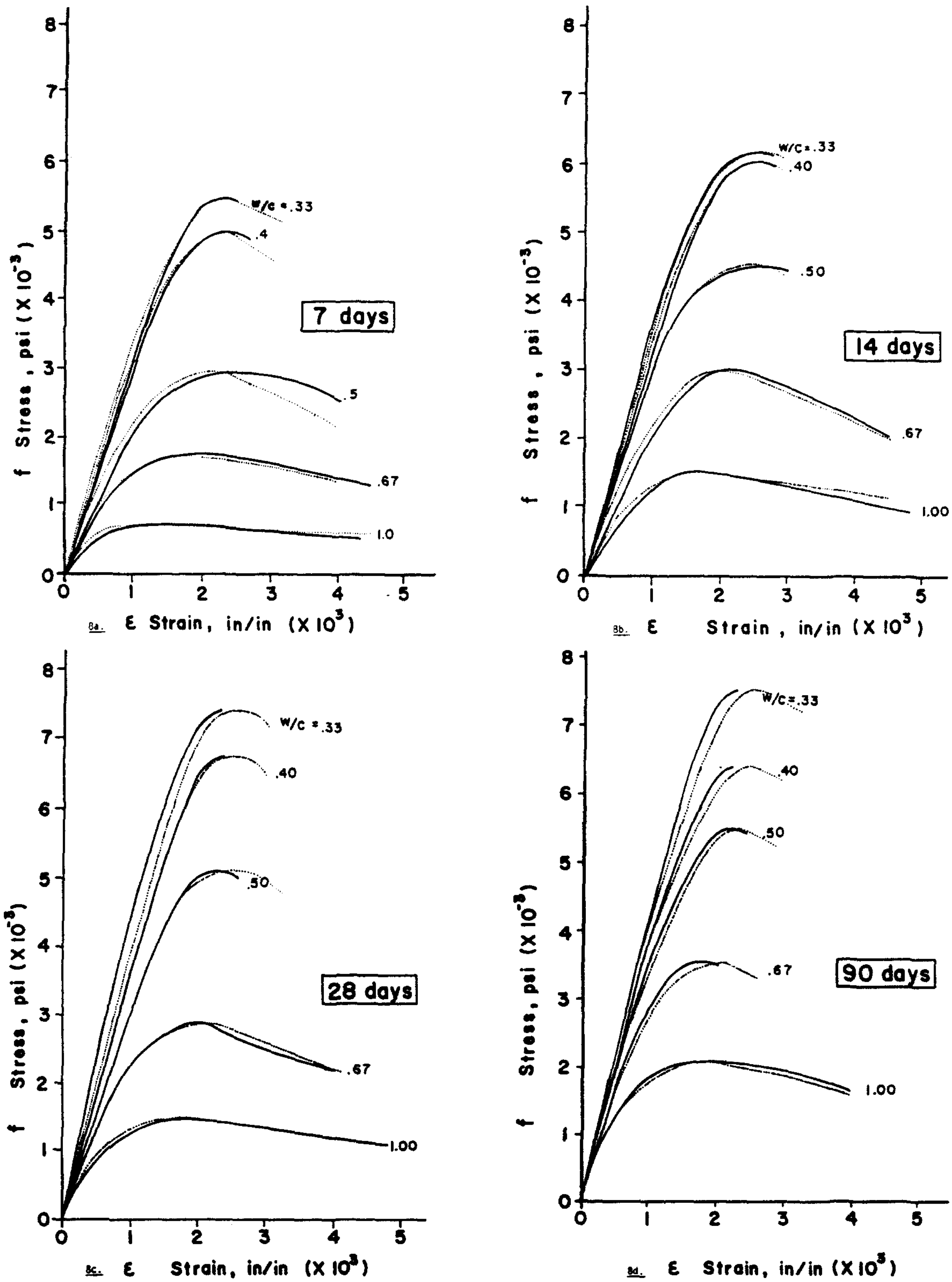


Fig. 7. Examples for the relationship between ε_o and f_o for light-weight as well as normal-weight concretes. The experimental data were taken from a paper by Watanabe (12).

experimentally obtained stress-strain diagrams by Smith and Young (9). It can be seen again that the presented formulas provide a good estimate of the complete stress-strain diagram of a concrete. As a matter of fact, the approximation of the proper combination of these formulas is better within the given wide ranges of water-cement ratio, compressive strength, and age than the approximations of other formulas recommended in the literature for the same purpose. (9) (13) - (16)

The stress-strain diagram for a concrete under short-term uniaxial tension is similar to the diagram produced by compression, except that the same curve is applicable for all the relative tensile stress-strain diagrams regardless of the strength. (17) It is probably more than coincidental that this single tension curve is very close to the compression curve presented in Figure 4 for pastes.



Figs. 8a through 8d. Comparison of experimental stress-strain diagrams of various concretes (continuous lines) to those plotted from Eq. 2) with Eqs. 3) and 6) (dot lines). The experimental curves were taken from Reference (11).

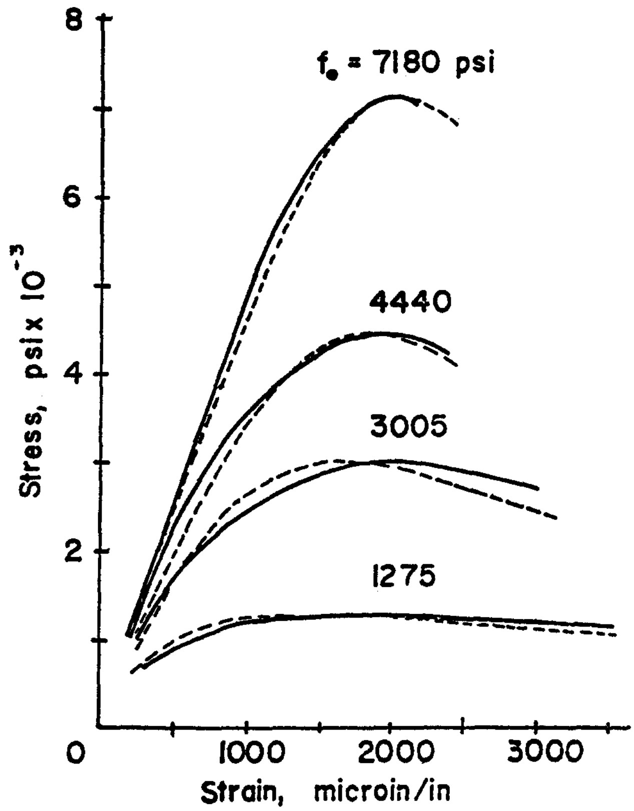


Fig. 9. Comparison of stress-strain diagrams of concrete cylinders (continuous lines) to those plotted from Eq. 2) with Eqs. 3) and 6) (dash lines). The experimental curves were taken from Reference (9).

Supplementary Remarks

(a) The area under any of the stress-strain diagrams calculated from the presented equations can be determined by numerical integration for which the computer programming is quite simple.

(b) Differentiation of Eq. 2) provides the E_t tangent modulus of elasticity of the concrete as follows:

$$E_t = df/d\varepsilon = \frac{n - 1 + (\varepsilon/\varepsilon_0)^n - n(\varepsilon/\varepsilon_0)^n}{[n - 1 + (\varepsilon/\varepsilon_0)^n]^2}\, nf_0/\varepsilon_0 =$$

$$= \frac{1 - (\varepsilon/\varepsilon_0)^n}{[n - 1 + (\varepsilon/\varepsilon_0)^n]^2}\, n(n - 1)f_0/\varepsilon_0 \qquad 7)$$

This becomes E at $\mathcal{E} = 0$, that is

$$E/E_o = \frac{n}{n - 1} \qquad 8)$$

which, with Eq. 3), provides the following relationship for normal-weight concretes:

$$E/E_o = 1 + \frac{2,500}{f_o} \qquad 9)$$

This means that E/E_o is 3.5 when f_o = 1,000 psi, and is 1.25 when f_o = 10,000 psi. These figures are in accordance with experimental data.

(c) Since $E_o = f_o/\mathcal{E}_o$, the combination of Eq. 9) with Eqs. 3) and 6) provides the following relationship between the compressive strength and initial modulus of elasticity of concrete:

$$E = \frac{10^4}{k} \frac{f_o + 2,500}{\sqrt[4]{f_o}} \qquad 10)$$

This equation fits reasonably well, within 2,000 and 10,000 psi compressive strength limits, the traditional empirical formula for normal-weight concretes:

$$E = K\sqrt{f_o} \qquad 11)$$

(d) It follows from Eq. 8) that

$$\mathcal{E}_o = \frac{nf_o}{(n - 1)E} \qquad 12)$$

Therefore, with the consideration of Eq. 3):

$$\mathcal{E}_o = \frac{1}{E}(f_o + 2,500) \qquad 13)$$

If E is given, or it can be estimated (18), Eq. 13) provides ε_0 in terms of the concrete strength. For instance, by substituting Eq. 11) for E:

$$\varepsilon_0 = \frac{1}{K}\left(\sqrt{f_0} + \frac{2,500}{\sqrt{f_0}}\right) \qquad 14)$$

The peculiarity of Eq. 14) is that the value of ε_0 decreases with the decrease of f_0 up to a point (which is in our case 2,500 psi), then it starts increasing so that as f_0 approaches zero, ε_0 approaches infinity. Since experimental data, such as shown in Figure 6, do not seem to show such an increase in ε_0 at low strengths, one can conclude that Eq. 14) may not hold for low strength normal-weight concretes because Eq. 11) may not hold for such concretes.

Conclusions

The proper combination of Eqs. 2), 3), and 6) seems suitable for the estimation of the complete stress-strain diagram of normal-weight concretes, made with a given aggregate and tested with a given procedure, either from the f_0 and ε_0 values, or solely from the values of f_0. As Figures 5a through 9 demonstrate, the approximation of these formulas is better within the given wide ranges of water-cement ratio, strength and age than the approximation of other formulas recommended in the literature for the same purpose. Mathematical analysis of Eqs. 1) through 6) provides additional relationships between the strength and deformation of a concrete.

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NOTATION

The letter symbols used in this paper and their definitions are as follows:

E	=	initial modulus of elasticity of the concrete,
E_o	=	secant modulus of elasticity at the f_o ultimate stress, that is, $E_o = f_o/\varepsilon_o$,
E_t	=	tangent modulus of elasticity at the ε strain, that is, $E_t = df/d\varepsilon$,
f	=	axial stress in the concrete specimen,
f_o	=	ultimate stress; in compression it means the cylinder strength,
k,K	=	experimental parameters,
ε	=	unit strain in concrete caused by the f stress, and
ε_o	=	unit strain in concrete at the f_o ultimate stress.